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transmissivity.py ×
1 import numpy as np
2 import matplotlib as mpl
3 from matplotlib import pyplot as plt
4
5 zlist = np.linspace(start=0, stop=90, num=90)
6 def Tp(z): 1个用法
7     z_rad = np.radians(z)
8     m = 1/np.cos(z_rad)
9     return np.exp(-0.1114*m)
10 def TK(z): 1个用法
11     z_rad = np.radians(z)
12     m = 1/(np.cos(z_rad)+0.50572*np.power(96.07995-z,-1.6364))
13     return np.exp(-0.1114 * m)
14 fig, ax = plt.subplots(figsize=(8,8*0.618), constrained_layout=True)
15 ax.plot(*args: zlist,[Tp(z) for z in zlist],c='black',ls='--',marker='o',lw=1,ms=5,label='Plane Parallel Atmosphere')
16 ax.plot(*args: zlist,[TK(z) for z in zlist],c='red',ls='-.',marker='s',lw=1,ms=5,label='Kasten and Young (1989)')
17 ax.axvline(x=60,c='black',ls=':')
18 ax.set_xlabel(xlabel: 'Zenith Distance [deg]',fontsize='xx-large')
19 ax.set_ylabel(ylabel: 'Transmissivity',fontsize='xx-large')
20 ax.xaxis.set_minor_locator(mpl.ticker.MultipleLocator(5))
21 ax.yaxis.set_minor_locator(mpl.ticker.MultipleLocator(0.05))
22 ax.legend(loc='best',fontsize='xx-large')
23 plt.tick_params(right='on', top='on', which='both')
24 ax.tick_params(which='major',length=5,width=1.5,labelsize='x-large')
25 ax.tick_params(which='minor',length=3)
26 plt.show()
27 fig.savefig(fname: 'Transmissivity.png', format='png', dpi=600, bbox_inches='tight', pad_inches=0.2)

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